

Forest-fire model with natural fire resistance

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Observations suggest that contemporary wildfire suppression practices in the United States have contributed to conditions that facilitate large, destructive fires. We introduce a forest-fire model with natural fire resistance that supports this theory. Fire resistance is defined with respect to the size and shape of clusters; the model yields power-law frequency-size distributions of model fires that are consistent with field observations in the United States, Canada, and Australia.

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I. INTRODUCTION

Responding to the risk to life and property posed by wildfires, many wildland management agencies in the United States and other developed nations have adopted aggressive fire suppression programs. Evidence suggests, however, that these efforts have actually produced conditions that facilitate large, dangerous fires [1–6]. We address this paradox by using simple models from the physics literature [7–10] to characterize the behavior of wildfires. Our model suggests that small early season fires, left to burn to natural extinction, contribute to a region’s natural fire resistance. By aggressively suppressing these fires, anthropogenic wildfire management can facilitate conditions that significantly increase the rate of large, destructive events.

In general, wildfires are part of a natural carbon cycle in regions where flora biomass decomposes slowly relative to its rate of accumulation, most notably in regions that experience distinct wet and dry seasons [1]. We model wildfire systems as complex systems in which clusters of flora form fuel networks whose connectivity depends on local biomass density, air temperature, fuel and air moisture content, and wind conditions. A region’s susceptibility or risk is related to the connectivity and breadth of the fuel network. Fires constitute a relaxation mechanism, occurring when human or natural agents provide a spark, for example a carelessly discarded cigarette or a lightning strike. Once a fire is started, it spreads through, and removes from the system, connected segments of the fuel network. When this network achieves a dense, highly connected state during the dry fire season, wildfire events closely fit a power-law frequency-size distribution (FSD),

$$N_{\text{fires}}(A) \propto A^{-b}, \quad (1)$$

where A is the burn area of the fire and the regional noncumulative slope (exponent) b has been measured between approximately $1.3 > b > 2$ [2,7,11]. Under these conditions, large fires occur with finite probability; the rate at which they occur is related to the exponent b .

In this paper, we introduce a variation of the Drossel-Schwabl forest-fire model (DSFFM) [10] that includes an

immunity dynamic. The model shows a relationship between the expected intensity of fires and the slope b of Eq. (1). We show that a region’s fire susceptibility is related to the rate at which fires propagate, specifically the rate at which the fire’s area increases, which is related to the shapes of the fuel clusters. The model can alternatively be interpreted with respect to pest infestations, disease, Ising models, and nearest-neighbor-propagated endemic processes in general [8]; for brevity, this paper focuses on the wildfire application. The model produces FSD of model fires with $1.2 < b < 1.74$, consistent with those observed in nature [7,11]. We further show that indiscriminately extinguishing early season fires can increase a region’s rate of large fires by compromising its natural fire resistance. Our model suggests that a progressive wildfire management program can mitigate wildfire risk by properly managing early season fires.

II. WILDFIRE SYSTEMS

Since the late 19th century, motivated by several particularly devastating fires and encouraged by some early successes, a philosophy of aggressive suppression has prevailed as the backbone of wildfire management in the United States [3]. Observations, however, suggest that by significantly extending fire return intervals (FRI), anthropogenic fire suppression permits fuel networks to grow dense and broadly connected and undermines the Darwinian advantage of fire resistant species. As fuel loads become more fire susceptible, particularly during the hottest and driest periods of the fire season, suppressing all ignition events becomes asymptotically impossible, and subsequent fires burn at very high temperatures. Fires in regions where small fuels are frequently removed by human or natural processes burn at lower temperatures and are significantly less destructive [3,4,6].

Minnich and Chou [2,5] specifically suggest a human connection to large wildfires in southern California, USA (SCA). They argue that by suppressing very small fires, during the relatively wet and cool part of the fire season, anthropogenic fire suppression in SCA has effectively shifted the bulk of the fire season into a regime of late season “extreme conditions.” They show that large fires in SCA occur much more frequently than in the adjacent region of Baja California Norte, Mexico (BCA), specifically that the slope b , in Eq. (1), is much steeper in BCA than SCA (see

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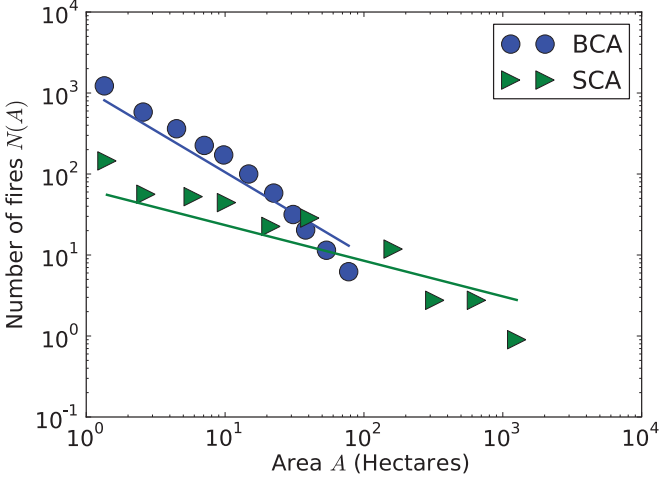


FIG. 1. (Color online) The frequency-size distributions of wildfires in southern California (SCA), green triangles, and northern Baja California Norte (BCA), blue circles. The published slopes of the log-binned distributions, $b_{SCA} = 0.44$ and $b_{BCA} = 1.02$ [2], are equivalent to unbinned, noncumulative slopes of $b_{SCA} = 1.44$ and $b_{BCA} = 2.02$, consistent with [11] and our simulations.

Fig. 1). They show that mean FRI are much longer in SCA than in the adjacent region of BCA, that fires in BCA burn earlier in the season, under cooler and higher relative humidity conditions, and that potentially large fires in BCA (during dry, windy conditions) are often constrained by burn patches where earlier fires have removed fuels. Because the two regions are contiguous and physically very similar, it is logical to attribute these distinctions to anthropogenic land management practices. Differences in fire suppression policies, namely that SCA aggressively suppresses wildfires and most fires in BCA burn to natural extinction, are especially suspect, although factors such as flora conservation programs, livestock grazing practices, and wildlife populations may also contribute [2,5].

III. SELF-ORGANIZED CRITICAL MODELING OF WILDFIRE SYSTEMS

The power-law behavior and nonlinear response to perturbations of wildfire systems, in this case the paradoxical apparent increased rate of large fires in response to fire suppression, suggests a relationship to a class of so-called self-organized critical (SOC) cellular automata (CA) models [12,13]. The Drossel-Schwabl forest-fire model (DSFFM) [10] has been broadly accepted, with some exceptions [14], as being a useful model of natural wildfire systems [7,11,15–21]. The standard DSFFM is defined as follows [10]

- (1) On a fixed-boundary $L \times L$ square grid, we select an element at random.
- (2) If that square is empty ($z = 0$), we plant a tree; the square becomes occupied ($z = 1$). If the square was already occupied, we do nothing.
- (3) With probability $1/f$, where f is a tuning parameter, we select a new square at random for a model lightning strike.
- (4) If that square is empty, do nothing; start over at (1); if that square is occupied, it ignites ($z = -1$).
- (5) All squares adjacent to a burning site ignite until the entire cluster is burning.

(6) The area of the fire k_f (the number of burning elements) is recorded.

(7) Burning squares return to an unoccupied state ($z = 0$); repeat from (1).

The DSFFM produces power-law FSD of model fires,

$$N_k \propto k_f^{-b}, \quad (2)$$

where N_k is the number of model fires involving k_f elements and the exponent (log-log slope) $b \approx 1$. Equation (2) is similar to Eq. (1), where $k_f = A$, with the understanding that values of A are typically measured from fire perimeters (and where the areas of unburned interior patches are not subtracted) and $N(A)$ are often binned. In contrast, N_k are simulated data, restricted to integer values and known exactly; k_f is explicitly the number of discrete elements, or “trees,” removed in a model fire.

To first order the DSFFM is widely accepted as behaving independently of lattice size, and it can be shown using simulations that the slope b over the power-law domain of the FSD is approximately independent of the ignition frequency f . In this study, we measure the slope of the standard DSFFM $b_{DSFFM} = 1.2$. Variations in the DSFFM that produce FSD with b steeper or shallower than b_{DSFFM} can be linked to physical conditions and processes that yield the range of b observed in natural systems. Minnich and Chou [2,5] further suggest that these variations can be emulated in land management policies to measurably mitigate, or exacerbate, wildfire risk.

Clar et al. [8] published a variation of the model that includes an immunity dynamic. They introduce a tuning parameter g and modify rule 5 such that, “A green tree [$z = 1$] becomes a burning tree [$z = -1$] with probability $(1 - g^n)$ if n of its nearest neighbors is burning. Otherwise it remains green.” The parameter g introduces a critical threshold $g_{th} = 1/2$ above which fires effectively do not occur (large fires are exponentially rare), and below which they recover the standard DSFFM with $b \approx 1$. This null result can also be demonstrated by a simple mean-field analysis in which the effective occupation probability of lattice sites that are nearest neighbors (NN) to burning elements is $\rho_{eff} = \rho(1 - g^{(n)})$, where ρ is the actual occupation probability of NN sites and $\langle n \rangle$ is the expected number of burning ($z = -1$) nearest neighbors. It follows that the expected number of trees removed by a fire k_f in a cluster of size k_c is

$$k_f = k_c(1 - g^{(n)}), \quad (3)$$

referring to the familiar mean-field analysis of the standard DSFFM [22],

$$N_f(k_f) = N_c(k_c)k_c, \quad (4)$$

where N_c , the number of clusters of size k_c , is a power law, Eq. (2), with slope $b_c = 1 + b_{DSFFM}$, and in the absence of immunity, $k_f = k_c$. Substituting Eqs. (2) and (3) into Eq. (4),

$$N_f(k_f) = \left(\frac{k_f}{1 - g^{(n)}} \right)^{1-b_c}. \quad (5)$$

To the extent that $g^{(n)}$ is independent of k_f , the effective distribution of occupied sites during a fire remains unchanged, and so the FSD of fires is not altered.

In this context, and in agreement with our own simulations, the value of b appears to be robust under changes to the rules that govern interactions between individual lattice sites. An alternative approach is to define an immunity dynamic with respect to information contained in the shape composition of the lattice. That is to say, a cluster of k discrete elements can assume a number of shapes, known as “lattice animals” [23]. In this context, a “shape” is any unique arrangement of occupied lattice sites that are NN contiguous. The state of a CA model can be described with respect to the collection of shapes represented on the lattice: its shape composition. Similarly, an individual cluster can be constructed from a basis of smaller shapes.

The circumference C of a cluster is defined as the number of discrete elements in its perimeter; a cluster’s perimeter consists of all empty sites that are NN to cluster member sites. For finite k , C is not well defined and can range from approximately $2k + 2 < C < \sqrt{2\pi k}$ for the linear and circular extreme cases, respectively [23]. In the context of wildfire models, and NN propagation CA models in general, this suggests that the rate at which a fire propagates through a cluster of k_c trees can vary significantly depending on the shape composition of the cluster relative to the epicenter of the model fire. With this in mind, we modify rule 5 of the DSFFM as follows:

(5a) All occupied sites adjacent to a burning site ignite.

(5b) The fire extinguishes completely with probability $P_q(k'_f)$, where k'_f is the current number of burning elements, and $k'_f \leq k_f \leq k_c$. Otherwise, with probability $1 - P_q(k'_f)$, repeat (5a).

For the scope of this work, we restrict ourselves to $P_q(k'_f)$ of the form

$$P_q(k'_f) = \epsilon p(k'_f), \quad (6)$$

where ϵ is a tuning parameter and $p(k'_f)$ is the functional model of immunity. Consistent with the observation that large fires are more difficult to control than small fires, we consider only $p(k'_f)$ that decrease with increasing k'_f .

Rules (5a) and (5b) produce two classes of fires and introduce a time scale. In the first class, a size $k_c = k_f$ cluster is selected and burns entirely. In the second, the fire survives through a set of intermediate sizes $k'_f < k_f$ and arrests according to Eq. (6) when $k'_f = k_f$. We update the fire (rule 5a) in parallel, before immunity is evaluated (rule 5b), so the rate at which the fire grows, $k'_f \equiv \frac{dk'_f}{dn}$, specifically the number of new elements added to a fire per simulation time step n , depends on the shape composition of its host cluster. Consequently, the number of times and values of k'_f at which Eq. (6) is evaluated, and therefore the cumulative probability that the fire persists, varies according to the shape of the cluster. For a fire triggered at the end of a linear cluster, $k'_f = 1, k'_f = 2, 3, 4, (n+1)$. For a fire in the middle of a linear cluster $k'_f = 2, k'_f = 3, 5, 7, (2n+1)$, and for a fire near the center of a large solid cluster, $k'_f = 4, 8, \dots, 4n$ and $k'_f = 5, 13, \dots, 1 + 2(n^2 + n)$. We evaluate immunity after the first generation of fire propagation where $1 \leq k'_f \leq 5$, so $0 \leq \epsilon < 5$.

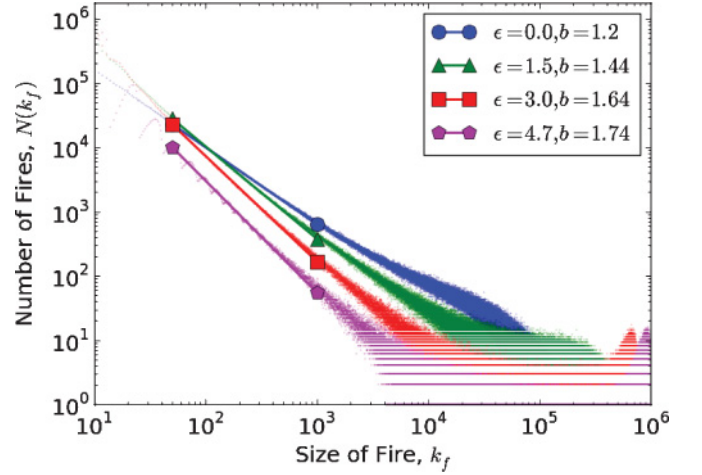


FIG. 2. (Color online) Frequency-size distributions, $N(k_f)$, and their least-squares fits for DSFFM-MFI simulations using Eq. (7). $\epsilon = 0, 1.5, 3.0, 4.7$ from top to bottom. We measure $1.2 < b < 1.74$, increasing approximately linearly with increasing ϵ (see Fig. 3).

IV. SIMULATION RESULTS

We conduct our simulations on an $L = 1024$ lattice with $f = 4096$; for Eq. (6), we use

$$P_q(k'_f) = \frac{\epsilon}{k'_f}. \quad (7)$$

We observe approximate power-law FSD with slopes $1.2 \leq b \leq 1.74$, increasing linearly as a function of increasing ϵ (see Figs. 2 and 3). The logarithms of the data are fitted to a line by a least-squares method; data for small and large k_f are omitted to exclude finite size anomalies.

We find that the model is qualitatively not sensitive to lattice size or sparking frequency, but does depend on the functional form of Eq. (6). For rapidly converging functions, such as exponential immunity,

$$P_q(k'_f) = \epsilon e^{-k'_f/\kappa}, \quad (8)$$

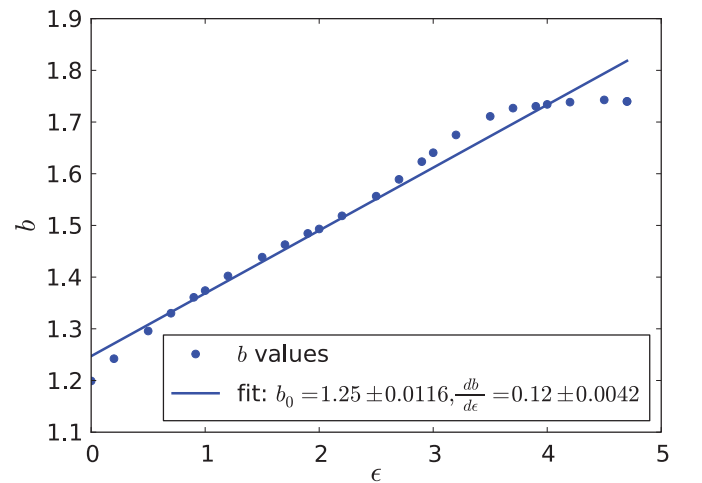


FIG. 3. (Color online) Least-squares best fit values for $b(\epsilon)$ from DSFFM-MFI simulation frequency-size distributions (blue circles). A linear least-squares fit (blue line) finds intercept $b_0 = 1.23 \pm 0.006$ and slope $db/d\epsilon = 0.13 \pm 0.0025$ for $0 < \epsilon < 4.0$.

where κ is a parameter, the FSD converges to the standard DSFFM distribution for $k_f \gg \kappa$. This suggests that different types of physical immunity, for example anthropogenic fire suppression and meteorological events, might affect fire dynamics very differently. A similar conclusion can be drawn for a step function model for fire suppression,

$$P_q = \Theta(k'_f - k_\theta), \quad (9)$$

where all fires are either suppressed after the first step if $k'_f < k_\theta$ or burn out of control (the entire cluster is removed). Simulations and a simple mean-field analysis of the model show that step-function immunity shifts the system to an elevated mean lattice occupation probability, but by itself does not change the slope of the FSD. This is as expected, since this type of step function immunity can be mapped to a lattice renormalization (effectively mapping to a smaller lattice) or emulated by increasing the ignition interval f .

V. MEAN-FIELD TREATMENT

In general, the the expected number of fires size k_f is

$$N_f(k_f) = N_{\text{quenched}}(k_f) + N_{\text{complete}}(k_f), \quad (10)$$

where N_{quenched} represents the number of fires that were extinguished and N_{complete} represents the number of fires in which an entire cluster was burned. Implementing rules (5a) and (5b) and inserting Eq. (6) into Eq. (10),

$$N_f(k_f) \propto \sum_{k_c > k_f}^{k_{\max}} N_c(k_c) k_c \left[\prod_{k'_f \in k_{\text{shapes}}}^{\max(k'_f < k_f)} (1 - P_q(k'_f)) \right] P_q(k_f) + N_c(k_f) k_f \left[\prod_{k'_f \in k_{\text{shapes}}}^{\max(k'_f < k_f)} (1 - P_q(k'_f)) \right], \quad (11)$$

where k_c and $N_c(k_c)$ are, respectively, the size (number of discrete elements) and number of clusters of size k_c , P_q is Eq. (6), and $k_{\max}$ is a maximum cluster size nominally related to the size of the lattice, $k_{\max} \approx L^2$. The first term represents the probability that a cluster size $k_c > k_f$ is selected, the fire survives through a series of sizes $k'_f < k_f$, and then arrests at $k'_f = k_f$, according to rules (5a) and (5b) and Eq. (6). The second term represents the probability that a size $k_c = k_f$ cluster is selected and burns completely. The cumulative probabilities of these outcomes depend on the discrete set of $k'_f \in k_{\text{shapes}}$ at which Eq. (6) is evaluated, which as discussed, depends on the shape composition of the cluster relative to the epicenter of the fire.

Substituting Eq. (7),

$$N_f(k_f) \propto \sum_{k_c > k_f}^{k_{\max}} N_c(k_c) k_c \left[\prod_{k'_f \in k_{\text{shapes}}}^{\max(k'_f < k_f)} \left(1 - \frac{\epsilon}{k'_f} \right) \right] \frac{\epsilon}{k_f} + N_c(k_f) k_f \left[\prod_{k'_f \in k_{\text{shapes}}}^{\max(k'_f < k_f)} \left(1 - \frac{\epsilon}{k'_f} \right) \right]. \quad (12)$$

Note that for the case $P_q = 0$, including $\epsilon = 0$ for Eq. (12), Eqs. (11) and (12) reduce to the familiar DSFFM mean-field solution, Eq. (4). In principle, values of k'_f can be determined

for specific shapes. Alternatively, particularly for large clusters for which many shapes are available, we assume the fractal relationship

$$k'_f = 1 + \rho_0 \alpha n^D, \quad (13)$$

and the fractal circumference of the shape, which determines a fire's rate of growth, is

$$C_{\text{frac}} = k'_f = \rho_0 \alpha D n^{D-1}, \quad (14)$$

where ρ_0 is the occupation probability of NN sites, α is a parameter, $n = 0, 1, 2, \dots, n_{\max}$ is the number of propagation steps or radius of the shape, and D is the fractal dimension of the shape. Linear shapes have fractal dimension $D = 1$, so $C_{\text{frac}} = \rho_0 \alpha$ and $k'_f = \rho_0 \alpha n$. For $n = 1$, these include rays, lines, and “L,” “T,” and cross shapes, for which $\alpha = 1, 2, 2, 3, 4$, respectively. For a circle or square, $D = 2$; circular geometry is satisfied when $\alpha = \pi$.

Assuming an isotropic and radially symmetric seed shape, we use $\alpha = 4$. For simplicity, we assume $\rho_0 = 1$, so the distributions of occupied and unoccupied sites within the extent of the shape are characterized by D and, to a lesser extent, α .

From simulations, we measure the FSD of clusters $N_c(k_c)$ to be a power law, Eq. (2), with slope $b_c = 2.2$, approximately constant for sufficiently large lattices. We also consider the further simplified case $b_c = 2.0$, which produces similar results. We substitute this into the first term of Eq. (12), convert the summation to an integral, and substitute Eq. (13), which yields

$$N_f(k_f) \propto \left(\frac{k_{\max}^{2-b_c} - k_f^{2-b_c}}{(2-b_c)} \right) \frac{\epsilon}{k_f} \prod_{n=1}^{(k_f-1)^{1/D}-1} \left(1 - \frac{\epsilon}{1+4n^D} \right) + k_f^{1-b_c} \prod_{n=1}^{(k_f-1)^{1/D}-1} \left(1 - \frac{\epsilon}{1+4n^D} \right), \quad (15)$$

and similarly for the approximation $b_c = 2.0$,

$$N_f(k_f) \propto \ln \left(\frac{k_{\max}}{k_f} \right) \frac{\epsilon}{k_f} \prod_{n=1}^{(k_f-1)^{1/D}-1} \left(1 - \frac{\epsilon}{1+4n^D} \right) + k_f^{1-b_c} \prod_{n=1}^{(k_f-1)^{1/D}-1} \left(1 - \frac{\epsilon}{1+4n^D} \right), \quad (16)$$

where $k_{\max} \approx L^2$, and presumably $1 < D < 2$. The upper values for the product functions are calculated by inverting Eq. (13) for $n = n(k'_f) - 1$, the second to final step of the fire's propagation. We numerically compute N_f for the values of k_f , according to Eq. (13), that correspond to sequential integer values of $n = 1, 2, 3, \dots, [n(k_f) - 1]$. We find that the scaling exponent b varies with both ϵ and D , and that the effectiveness of ϵ , specifically the range of b as a function of ϵ , diminishes with increasing D , as can be seen in Figs. 4, 5, and 6. We find $1.2 < b < 3.3$ within the parameter domain, $0.9 < D < 2.5$, and $0 < \epsilon < 4.9$, recognizing of course that $1 > D > 2$ is arguably nonphysical. As expected, we recover the standard DSFFM behavior, $b \approx 1.2$, for all ϵ when $D > 1.4$ —where

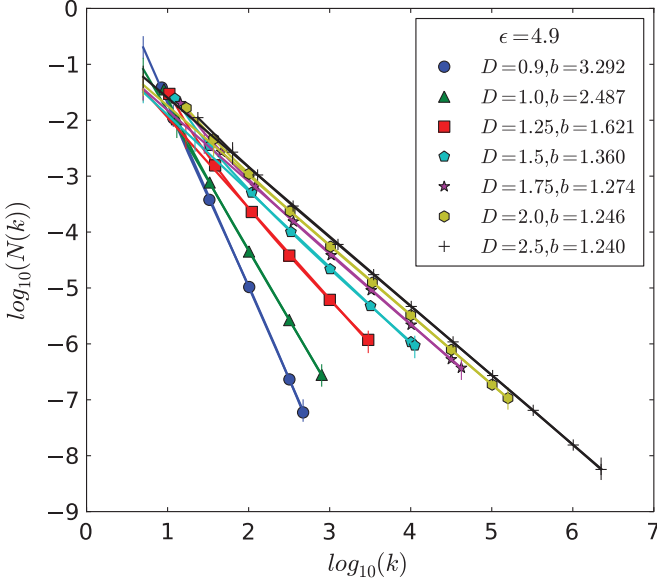


FIG. 4. (Color online) Frequency-size distributions, $N(k_f)$, for the mean-field treatment of DSFFM-MFI using Eq. (7). In this plot, $b_c = 2.2$, $\epsilon = 4.9$, and $0.9 < D < 2.5$, increasing from bottom (blue circles) to top (black “plus” markers). We measure $1.24 < b < 3.3$; the steepest distribution, $b = 3.3$, is associated with the smallest fractal dimension $D = 0.9$ (blue circles). The scaling exponent b decreases systematically as a function of increasing D .

rapid fire growth nullifies the immunity dynamic. These results are consistent with results from our simulations and show that the steeper slopes b result from the immunity dynamic as opposed to an artifact of the simulation such as characteristic (lattice size) events. A similar, but less pronounced, result can

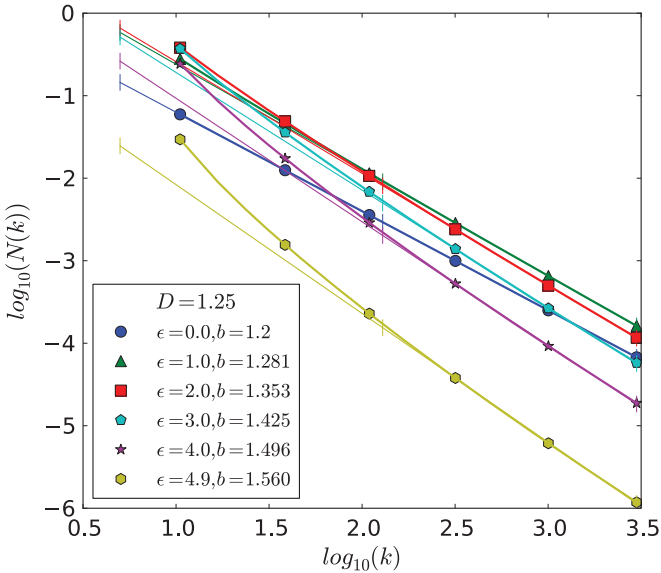


FIG. 5. (Color online) Frequency-size distributions, $N(k_f)$, for the mean-field treatment of DSFFM-MFI using Eq. (7). In this plot, $b_c = 2.2$ and $D = 1.25$. Blue circles, green triangles, red squares, cyan pentagons, purple stars, and gold hexagons represent $\epsilon = 0.0, 1.0, 2.0, 3.0, 4.0$, and 4.9 , respectively. We measure $1.2 < b < 1.56$, increasing approximately linearly with increasing ϵ .

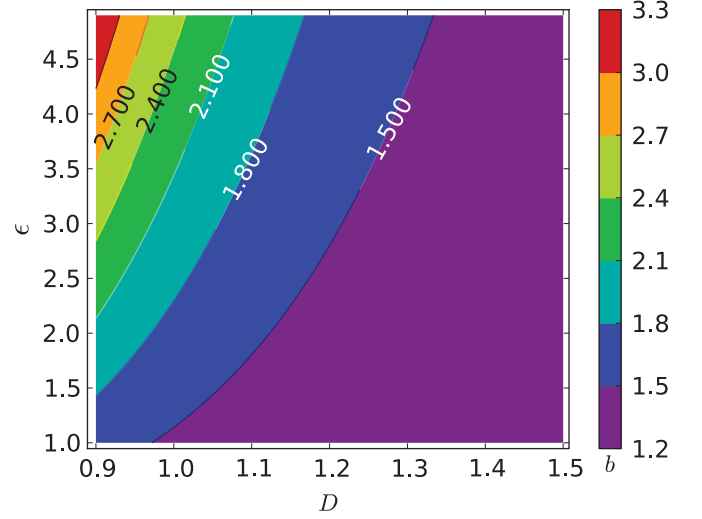


FIG. 6. (Color online) Contour plot of $b(D, \epsilon)$, as measured from the mean-field treatment of DSFFM-MFI. The scaling exponent b varies systematically with both ϵ and D , decreasing from $b = 3.3$ (top left) to $b = 1.2$ (bottom right). Also, the range of b as a function of ϵ diminishes for higher values of D .

be produced by assuming large, solid clusters ($D = 2$) and varying $0 < \rho_0 \alpha < 4$.

VI. CONCLUSION

We introduce an immunity variation of the DSFFM that produces approximate power law frequency-size distributions (FSD) of model fire events. Our numerical simulations produce slopes in the range $1.2 < b < 1.74$ as a function of a tuning parameter ϵ , which emulates regional fire susceptibility, or equivalently, the expected intensity of a fire. This is similar to the range of slopes observed in wildfire data from various regions of the United States, Canada, and Australia and is consistent with the observation that b values are steeper in the southeastern United States than in the much drier southwest [7,11].

We also introduce a mean-field analysis of the model that produces power-law FSD with slopes b spanning values observed in nature and in our simulations. As shown in Figs. 4, 5, and 6, the scaling exponent b varies systematically with both the cluster’s fractal dimension D , which parameterizes the cluster’s shape, and the parameter ϵ . The range of b , as a function of ϵ , diminishes for higher values of D , which is consistent with expectations as per Eq. (14). Our mean-field analysis suggests that while the size distribution of clusters does not change significantly with ϵ (b_c is approximately constant), the distribution of cluster shapes—or at least the effective shapes under the influence of immunity—varies substantially.

The long-range interactions between elements of large fires, resulting from Eq. (6), can be physically justified with respect to long-range transport of embers and correlated airflow through the burn area—phenomena for which large fires are notorious. Alternatively, Eq. (6) can be interpreted to represent the probability that a meteorological event of sufficient magnitude quenches the fire; rainfall events and droughts, for

example, are known to exhibit frequency-magnitude scaling [24].

We show that while $1/k'_f$ type immunity, Eq. (7), produces power-law FSD with $b > b_{\text{DSFFM}}$, immunity functions that converge more rapidly, Eqs. (8) and (9), produce distributions with $b = b_{\text{DSFFM}}$. Observations suggest that such rapidly converging immunities are not only ineffectual, but can actually undermine more effective, $1/k'_f$ type perhaps, fire resistance mechanisms by prematurely extinguishing early season fires when they are very small.

This appears to be the case in SCA compared to BCA. In BCA, frequent early season mild intensity, high ϵ fires appear to significantly cull fuel supplies before extinguishing naturally. These fires are aggressively suppressed in SCA, both forcing fires to burn during the later, higher risk part of the fire season and extending FRI by a factor of approximately 7 [2].

Extended FRI have been correlated with increased fuel loads and high intensity fires [3]. This has traditionally been described with respect to the *density* ρ of the local fuel load, but a more accurate description is that extended FRI facilitate increased *connectivity* of the fuel network. More specifically, our model suggests that the fractal dimension D of the fuel network tends to increase with increasing FRI. Conversely, small early season fires tend to reduce D by removing connective fuel segments. It follows that a progressive fire management program can mitigate the rate of large, destructive wildfires by properly managing early season fires.

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